Unlocking Flexibility Potential through DVFS-controlled Data Center Clusters in Unit Commitment

***Abstract*—Data center clusters (DCCs) have become recognized as significant energy consumers in electrical power system operations. Workload balancing for DCCs, implemented by dynamic voltage and frequency scaling (DVFS) control schemes, allows for providing flexibility potential capability and contributes to maintaining the security of scheduling results through Unit Commitment (UC). To achieve this, the flexibility potential of DCCs under different workload conditions is analyzed. Building on this analysis, a new set of physical constraints representing workload balancing across multiple DVFS-controlled DCCs is proposed, and related linearized constraints is extended through McCormick Envelopes method. Furthermore, these intractable constraints are incorporated into UC, and an enhanced UC (EUC) model, which considers the flexibility potential offered by DVFS-controlled DCCs, is developed. Several experiments are conducted on the IEEE 6 bus and 118 bus test systems to assess the validity and scalability of the proposed models.**

***Index Terms*—Data center clusters, Flexibility Capability, Enhanced Unit Commitment, McCormick Envelopes**

I. Introduction

As the increase in various artificial intelligence technologies and demand energy consumption, the operation of electrical power systems are becoming diverse and uncontrollable, and they are facing challenges in provide sufficient flexible adjustment resources against emergence events [1], [2]. In such circumstances, additional flexible mechanisms such as demand response, virtual Power plant, and energy storage systems, etc., needs to be prioritized. Nevertheless, as a significant energy consumer, DCCs have not been fully utilized in the flexibility potential of power system operations. To this end, it it essential to investigate the flexible adjustment characteristic of DCCs, and to explore the potential of DCCs in providing flexible adjustment resources for the economic and reliable operation of electrical power systems.

Recent literature reported several studies on the DCCs flexibility modelling and its application in electrical power systems operation. Specially, Ref. [3] carried out a Quality-of-service-based cost function to describe the workload shifting and server utilisation of DCCs, and extended a distributed dispatch framework with considering DCCs flexibility. Ref. [1] designed a mix-integer programming model to achieve the the optimal demand management of DCCs. Ref. [2] analysed the progressive load characteristics in DCCs, and proposed two optimization models to estimate the optimal capability planning of storage systems in DCCs and design parameters optimization. Ref. [4] explored the statistical feasibility of DDCs and hydrogen resources in a microgrid context, and carried out a data-driven statistical-feasibility-based robust rolling optimization framework to analysis the optimal operation of DDCs and hydrogen storage systems under complex uncertain factors. Ref. [5] further analysed the potential capability of DDCs for regulating computational workloads while reducing carbon emissions, and constructed a emission-aware coordinated dispatch method of multi-regional microgrid with DDCs. The aforementioned studies primarily emphasise the flexibility capabilities of DCCs within electrical power system operations. However, they often overlook the intricate physical characteristics of DCCs, particularly in relation to workload shifting dynamics and the implementation of advanced high-evolution control strategies.

The workloads in DCCs can be broadly classified into three categories, [1], $i)$ $compute\text{-}intensive\ workloads$, which include high-performance computing, machine learning model training, and commercial application services. $ii)$ $storage\text{-}intensive\ workloads$, which are employed for tasks such as regular file backups, archival systems, and relational database applications. $iii)$ $network\text{-}intensive\ workloads$, encompassing activities like website hosting, content delivery networks, and network traffic management. Notably, $compute\text{-}intensive$ and $storage\text{-}intensive$ workloads dominate in DCCs, collectively accounting for approximately 70% of the total workload. This dominance is driven by the growing demand for high-performance computing and data-driven techniques. In contrast, $network\text{-}intensive\ workloads$ are typically less predictable and highly time-sensitive, requiring immediate processing to ensure seamless operation. Since the former two workloads are more predictable and flexible, the peak workload requirements of the former two workloads can be effectively shifted or local workload delay responsed through DVFS, [1], [2], [6], to achieve spatio-temporal workload balancing and total energy management.

In addition to the energy management process from DCCs using DVFS control schemes, additional physical constraints concerning task-size bandwidth and time-tolerant workload types must be taken into account, as they directly influence the feasibility of workload shifting between multiple DCC servers and their operational logic. For instance, excessively shifting time-sensitive workloads, particularly network-intensive ones, between DCCs may lead to potential risks such as data congestion and latency due to limited transmission bandwidth, thereby compromising the ability to meet actual real-time demands. To reflect this issue, Ref. [7] investigated a queuing model implemented in cloud-based DCCs using DVFS load-dependent strategies, and highlighted its significant on energy efficiency and workload management. Ref. [8] analysed the response characteristics of various time-tolerant tasks on workload completion time and total energy consumption, and proposed a dual-timescale collaborative optimization model for multi-DCC systems. Ref. [9] explored the impact of batch workload scheduling in DCCs on the energy consumption, and developed a day-ahead emission-aware stochastic optimization model for microgrid energy planning and generation management, Ref. [10] extended a data-driven distributionally robust optimization framework for low-carbon DCC systems, addressing multi-tasking responses and uncertainties in renewable energy integration. Ref. [6] proposed a workload-aware robust DVFS technique, termed as GearDVFS, to address concurrency challenges in mobile-driven workloads. Ref. [11] provided a comprehensive survey of efficient task scheduling and resource allocation methodologies in multi-DCC networks. Ref. [12] proposed a distributed shortest remaining time first (SRTF) scheduling model for DCCs, designed to minimise flow completion time while ensuring flow-size availability. Ref. [13] constructed a machine-learning-based scheduling flow detection and packet-switched optical network (PSON) architecture with centralized control for intra-DCC connectivity and bath workload allocation. These studies comprehensively address multi-task assignments and flow-size bandwidth limitations across different DCCs, further developing various task allocation models and analysing the energy consumption of received computing tasks. However, the potential interaction mechanisms between workload processes, particularly those involving multi-task shifting across multiple DCCs and their role in meeting electrical flexibility requirements, remain underexplored.

Driven by this motivation, this paper analyses the flexibility offered by DVFS-controlled DCCs and integrates this capability into UC studies. A key aspect of this work is the physical modeling of the interaction mechanism between DCC energy consumption and workload balancing strategies managed by DVFS. Building upon this foundation, additional restrictions such as bandwidth limitations and time-delay requirements for various workload types are reshaped into corresponding physical constraints and linearized reformulated. In contrast to previous studies that focused on workload shifting strategies or energy consumption reduction, this research provides a comprehensive analysis of the coordinated optimization between power system operations and the DCC workload execution process. Moreover, the multi-timescale interactions between active response characteristic of DCCs through workload shifting and grid-side demand fluctuation is modeled in detail, considering both grid-side requirements and specific implementation details of workload balancing processes.

In summarize, the main contributions of this paper are summarized: i) A DVFS-controlled physical constraints is developed to characteristic the flexibility attributes of DCCs workloads. Furthermore, the proposed constraints are linearized using McCormick Envelopes to enhance computational tractability in UC studies. ii) A EUC model is formulated to capture the dynamic flexible response of DCCs. The model incorporates a new coordination mechanism to address multi-timescale interactions between workload shifting in DCCs and grid-side demand fluctuations.

The remainder of this paper is organized as follows. Section II introduces the proposed scheduling framework, which integrates the flexibility potential of DVFS-controlled DCCs. Section III develops a new set of physical constraints to model DCC workload shifting and local energy consumption requirements. Additionally, the McCormick Envelope method is employed to linearized the nonlinear constraints, enabling the estimation of DCC flexibility potential. Section IV incorporates the flexibility potential of DVFS-controlled DCCs into UC and formulates the proposed EUC model in detail. Section V presents numerical experiments conducted on benchmark test systems to validate the effectiveness and scalability of the proposed model. Finally, Section VI summaries the key findings and insights derived from the case studies.

II. Reformulated Constraints for DVFS-controlled DCCs

This section presents a coordinated scheduling framework to integrate the flexible response capabilities of DCCs into electrical power system operations. Recognising the difference in scheduling timescales (hourly for grid-side operations versus seconds for DCCs workload management), the framework is structured into two interconnected parts. PART-1 encompasses the grid-side scheduling model, while PART-2 represents the DCCs workload management model.

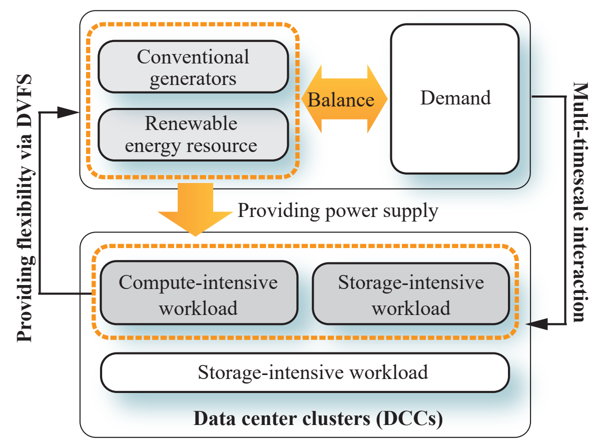

Fig. 1. The proposed scheduling framework in this work

The power consumption of DCCs is primarily influenced by two components: static energy and dynamic energy. The static energy term, often referred to as idle power, arises mainly from the leakage current in the CPU core circuits. In contrast, the dynamic energy term is predominantly determined by the workload ratio λ_{dcc} and the individual service level μ_{dcc}. As such, the total power consumption of DCCs can be expressed as follows, [6],

$$p_{dcc,i} = p^{idle}_{dcc,i} + p^{dyn}_{dcc,i}, \qquad \forall i \in \mathcal{I} \tag{1a}$$
$$p^{idle}_{dcc,i} = v_{dcc,i} i_{leak}, \qquad \forall i \in \mathcal{I} \tag{1b}$$
$$p^{dyn}_{dcc,i} = \left(C_{dcc,i} f_{dcc,i} v^2_{dcc,i}\right) \lambda_{dcc,i} \mu^{-1}_{dcc,i}, \qquad \forall i \in \mathcal{I} \tag{1c}$$

where i and $\mathcal{I}$ represent the index and set of DCCs. $p_{dcc,i}$ denotes the total power consumption of individual DCC i. $p^{idle}_{dcc,i}$ and $p^{dyn}_{dcc,i}$ represent the static (idle) and dynamic power consumption of DCC i, respectively. $C_{dcc,i}$ defines the constant representing the effective capacitance of DCC i. $f_{dcc,i}$ and $v_{dcc,i}$ denote the operational frequency and supply voltage of DCC i, respectively.

Based on the above physical working principles, the set of associated security constraints representing DCCs workflows can be extended,

$$
\begin{aligned}
&p_{dcc,i} \geq p_{dcc,i}^{idle} + p_{dcc,i}^{dyn}, && \forall i \in \mathcal{I} && \text{(2a)}\\
&p_{dcc,i}^{idle} \geq \lfloor v_{dcc,i} \rfloor i_{leak}, && \forall i \in \mathcal{I} && \text{(2b)}\\
&p_{dcc,i}^{idle} \leq \lceil v_{dcc,i} \rceil i_{leak}, && \forall i \in \mathcal{I} && \text{(2c)}\\
&p_{dcc,i}^{dyn} \geq \lfloor (C_{dcc,i} f_{dcc,i} v_{dcc,i}^2) \rfloor \lambda_{dcc,i} \mu_{dcc,i}^{-1}, && \forall i \in \mathcal{I} && \text{(2d)}\\
&p_{dcc,i}^{dyn} \leq \lceil (C_{dcc,i} f_{dcc,i} v_{dcc,i}^2) \rceil \lambda_{dcc,i} \mu_{dcc,i}^{-1}, && \forall i \in \mathcal{I} && \text{(2e)}
\end{aligned}
$$

where $\lceil \circ \rceil$ and $\lfloor \circ \rfloor$ represent pre-determined upper and lower bounds for the decision variable $\circ$. $\lambda_{dcc,i}$ describes the workload balancing process for the DCC i. The feasibility of workload balancing for compute-intensive and storage-intensive workloads hinges on users' latency tolerance for batch workloads and the capacity to shift workloads among multi DCCs. When users accept tolerate higher latency, the execution of workloads can be dynamically adjusted across multi DCCs. By optimizing these workload execution patterns, the overall energy consumption of DCCs can be optimized through DVFS and sequentially reduced, thereby enhancing their demand response capabilities for power system operations.

Next, to demonstrate the benefits of DVFS-controlled DCCs for achieving smooth energy demand and proactive workload response, additional concise constraints concerning workload balancing processes among multi DCCs are constructed,

$$
\begin{aligned}
&p_{dcc,i} \geq p_{dcc,i}^{idle} + p_{dcc,i}^{dyn} + \Delta p_{dcc,i}^{bal}, && \forall i \in \mathcal{I} && \text{(3a)}\\
&\Delta p_{dcc,i}^{bal} = \phi_{dcc,i}^{dvfs} \big(\lambda_{dcc,i}^{ini} + \Delta\lambda_{dcc,i}\big) \mu_{dcc,i}^{-1}, && \forall i \in \mathcal{I} && \text{(3b)}\\
&\phi_{dcc,i}^{dvfs} = C_{dcc,i} f_{dcc,i} v_{dcc,i}^2, && \forall i \in \mathcal{I} && \text{(3c)}\\
&\lfloor f_{dcc,i} \rfloor \leq f_{dcc,i} \leq \lceil f_{dcc,i} \rceil, && \forall i \in \mathcal{I} && \text{(3d)}\\
&\lfloor v_{dcc,i} \rfloor \leq v_{dcc,i} \leq \lceil v_{dcc,i} \rceil, && \forall i \in \mathcal{I} && \text{(3e)}
\end{aligned}
$$

where $\Delta p_{dcc,i}^{bal}$ represents the change in the power consumption for DCC i due to workload balancing, $\lambda_{dcc,i}^{ini}$ represents the initial workload, and $\Delta\lambda_{dcc,i}$ represents the adjusted workload. The term $\phi_{dcc,i}^{dvfs}$ represents the power regulation process for the DCC i through DVFS technique, By incorporating the workload balancing dynamics defined in (3a)-(3c) into (2b) and (2c), the form of DVFS-controlled DCCs constraints can be established.

The value of $\lambda_{dcc,i}^{bal}$ primarily depends on users' tolerance for computational workload response time. Generally speaking, greater tolerance allows for a higher $\lambda_{dcc,i}^{bal}$, thereby enhancing the workload balancing capability of DCCs and enabling more flexible adjustment of workloads across clusters. As such, the performance of workload balancing processes can be estimated through the metric of average workload queuing delay duration, $i.e.$, $T_q = 1/(v_{dcc,i} - \lambda_{dcc,i})$. To address this issue, the following lists the relate binding constraints concerning workload balancing processes,

$$
\begin{aligned}
&T_q \leq T_{tol}^{-1} : T_q = 1/(v_{dcc,i} - \lambda_{dcc,i}), && && \text{(4a)}\\
&\Rightarrow \lambda_{dcc,i}^{ini} + \lambda_{dcc,i}^{dyn} + \Delta\lambda_{dcc,i}^{bal} \leq v_{dcc,i} - T_{tol}^{-1}, && \forall i \in \mathcal{I} && \text{(4b)}\\
&\Rightarrow \Delta\lambda_{dcc,i}^{bal} \leq v_{dcc,i} - T_{tol}^{-1} - (\lambda_{dcc,i}^{ini} + \lambda_{dcc,i}^{dyn}), && \forall i \in \mathcal{I} && \text{(4c)}\\
&\Delta\lambda_{dcc,i}^{bal} \geq 0, \Delta\lambda_{dcc,i}^{bal} \in \mathbb{R}, && \forall i \in \mathcal{I} && \text{(4d)}
\end{aligned}
$$

where T_{tol} represents the predefined threshold for the average workload queuing delay duration.

It is worth to noting that constraints (3a)-(3c) remain elusive due to the presence of product terms involving continuous variables, $ie.$, $\phi_{dcc,i}^{dvfs}$ and $\Delta\lambda_{dcc,i}$. To address this, the McCormick envelope method is introduced to linear the constructed constraints mentioned before. More detailed, let $\iota_{dcc,i} = \phi_{dcc,i}^{dvfs} \times \Delta\lambda_{dcc,i}$, and additionally introduce two auxiliary variables $\varsigma_{dcc,i}^{up}, \varsigma_{dcc,i}^{dw}$ where $\varsigma_{dcc,i}^{up} = 1/2(\phi_{dcc,i}^{dvfs} + \Delta\lambda_{dcc,i})$, $\varsigma_{dcc,i}^{dw} = 1/2(\phi_{dcc,i}^{dvfs} - \Delta\lambda_{dcc,i})$, accordingly, the constraint (3b) can be reformatted as its approximated linear form as below,

$$
\begin{aligned}
&\Delta p_{dcc,i}^{bal} = \phi_{dcc,i}^{dvfs} \lambda_{dcc,i}^{ini} \mu_{dcc,i}^{-1} + \\
&\qquad \iota_{dcc,i} \mu_{dcc,i}^{-1}, \iota_{dcc,i} = \phi_{dcc,i}^{dvfs} \Delta\lambda_{dcc,i}, && \forall i \in \mathcal{I} && \text{(5a)}\\
&\iota_{dcc,i} \geq (\varsigma_{dcc,i}^{up})^2 - (\varsigma_{dcc,i}^{dw})^2, && \forall i \in \mathcal{I} && \text{(5b)}\\
&\varsigma_{dcc,i}^{up} \geq 0, \varsigma_{dcc,i}^{up} \in \mathbb{R}, \varsigma_{dcc,i}^{dw} \in \mathbb{R}, && \forall i \in \mathcal{I} && \text{(5c)}
\end{aligned}
$$

In (4b), the standard square terms of $\langle \varsigma_{dcc,i}^{up}, \varsigma_{dcc,i}^{dw} \rangle$ can be further reshaped as a Special-Ordered-Sets-of-type-2 (SOS2) constraint, which MILP solver could accurately identify and compile them. For simplification, let $\langle \varsigma_{dcc,i,z}^{up}, \varsigma_{dcc,i,z}^{dw} \rangle$, respectively, denote the piecewise block of related continuous variables $\langle \varsigma_{dcc,i}^{up}, \varsigma_{dcc,i}^{dw} \rangle$, hence, (4b) can be reformulated as below,

$$
\begin{aligned}
&\iota_{dcc,i} \geq \sum\nolimits_{z\in\mathcal{Z}} \Im(\varsigma_{dcc,i,z}^{up}) \varpi_{dcc,i,z}^{up} - \\
&\qquad \sum\nolimits_{z\in\mathcal{Z}} \Im(\varsigma_{dcc,i,z}^{dw}) \varpi_{dcc,i,z}^{dw}, && \forall i \in \mathcal{I} && \text{(6a)}\\
&\varsigma_{dcc,i}^{up} = \sum\nolimits_{z\in\mathcal{Z}} \varsigma_{dcc,i,z}^{up} \varpi_{dcc,i,z}^{up}, && \forall i \in \mathcal{I} && \text{(6b)}\\
&\varsigma_{dcc,i}^{dw} = \sum\nolimits_{z\in\mathcal{Z}} \varsigma_{dcc,i,z}^{dw} \varpi_{dcc,i,z}^{dw}, && \forall i \in \mathcal{I} && \text{(6c)}\\
&\sum\nolimits_{z\in\mathcal{Z}} \varsigma_{dcc,i,z}^{up} = 1, \sum\nolimits_{z\in\mathcal{Z}} \varsigma_{dcc,i,z}^{dw} = 1, && \forall i \in \mathcal{I} && \text{(6d)}\\
&\Im(\varsigma_{dcc,i,z}^{up}) = (\varsigma_{dcc,i,z}^{up})^2, \forall z \in \mathcal{Z}, && \forall i \in \mathcal{I} && \text{(6e)}\\
&\Im(\varsigma_{dcc,i,z}^{dw}) = (\varsigma_{dcc,i,z}^{dw})^2, \forall z \in \mathcal{Z}, && \forall i \in \mathcal{I} && \text{(6f)}\\
&\varsigma_{dcc,i,z}^{up} \geq 0, \varsigma_{dcc,i,z}^{dw} \geq 0, \forall z \in \mathcal{Z}, && \forall i \in \mathcal{I} && \text{(6g)}\\
&\varpi_{dcc,i,z}^{up} \in \{0,1\}, \varpi_{dcc,i,z}^{dw} \in \{0,1\}, \forall z \in \mathcal{Z}, && \forall i \in \mathcal{I} && \text{(6h)}
\end{aligned}
$$

Similarly, constraint (3c) can be approximated as a set of SOS2 constraints using the McCormick Envelope method. The detailed mathematical formulations for this approximation are provided in Appendix A. By combining these previously mentioned constraints, a comprehensive set of physical constraints describing workload power consumption can be effectively established.

III. Refined Workload Balancing Cosntraints among Multiple DCCs

As previously discussed, $\lambda_{dcc,i}$ serves as an indicator for the workload allocation process on individual DCC devices. This process is achieved by regulating the operational frequency and supply voltage parameters of each individual DCC. Nevertheless, the overall workload requirements remain constant, even though the specific workload assigned to individual DCCs might be adjusted (increased or decreased). To reflect this characteristic, the sum of $\lambda_{dcc,i}$ for all workloads hosted on a specific DCC device must be restricted as follows:

$$\sum\nolimits_{i\in\mathcal{I}}(\Delta\lambda_{dcc,i}+\lambda_{dcc,i}^{ini})=\sum\nolimits_{i\in\mathcal{I}}\lambda_{dcc,i}^{ini},\quad \forall i\in\mathcal{I}$$
$$\Rightarrow\sum\nolimits_{i\in\mathcal{I}}\Delta\lambda_{dcc,i}=0,\Delta\lambda_{dcc,i}\in\mathbb{R}\quad\forall i\in\mathcal{I}\tag{7a}$$

Given that compute workload types differ across DCCs, and not all workloads can be freely transferable to other DCC device, where this transferability is closely dependent on the current workload type (specifically, only $compute\text{-}intensive$ and $storage\text{-}intensive$ types are generally transferable, while $network\text{-}intensive$ type may not). This necessitates evaluating each DCC's potential to engage in workload balancing and establishing the following constraint:

$$\lfloor\Delta\lambda_{dcc,i}\rfloor\le\Delta\lambda_{dcc,i}\le\lceil\Delta\lambda_{dcc,i}\rceil,\quad\forall i\in\mathcal{I}\tag{8a}$$

where $\lceil\Delta\lambda_{dcc,i}\rceil$ and $\lfloor\Delta\lambda_{dcc,i}\rfloor$, respectively, denotes the upper bound and lower bound of available adjustable potential during workload balancing processes.

IV. Improved unit commitment through DVFS-controlled DCCs

In order to depict the flexibility potential provided by DDCs during the entire scheduling process, a refined coordination mechanism between active workload balancing hosted by multiple DCCs and electrical scheduling process is designed in this section, then a complete mathematical formulations concerning proposed scheduling model is extended.

$$\sum\nolimits_{t\in\mathcal{T}}\sum\nolimits_{g\in\mathcal{G}}[C_{su,g}u_{g,t}+C_{sd,g}v_{g,t}]+\tag{9a}$$
$$\sum\nolimits_{s\in\mathcal{S}}\sum\nolimits_{t\in\mathcal{T}}\Big[\sum\nolimits_{g\in\mathcal{G}}\Big(\sum\nolimits_{z\in\{1,2,3\}}\alpha_g^{(z)}p_{g,s,t}^{(z)}+x_{g,t}\beta_g^{(z)}\Big)+$$
$$\Big(\sum\nolimits_{w\in\mathcal{W}}C_{shed,w}\Delta p_{w,s,t}\Big)+\Big(\sum\nolimits_{d\in\mathcal{D}}C_{voll,d}\Delta p_{d,s,t}\Big)\Big]$$

$$s.t.,u_{g,t}-v_{g,t}=x_{g,t-1}-x_{g,t},\quad\forall g\in\mathcal{G},\forall t\in\mathcal{T}\tag{9b}$$

$$x_{g,t}-x_{g,t-1}\le u_{g,t},\quad\forall g\in\mathcal{G},\forall t\in\mathcal{T}\tag{9c}$$

$$\sum\nolimits_{m\in\mathcal{M}(\omega_g^{dt}-1,t)}(1-u_{g,m})\ge\omega_g^{dt}(u_{g,t}-u_{g,t-1}),$$
$$\forall g\in\mathcal{G},\forall t\in\mathcal{T}\tag{9d}$$

$$\sum\nolimits_{m\in\mathcal{M}(\omega_g^{ut}-1,t)}u_{g,m}\ge\omega_g^{ut}(u_{g,t-1}-u_{g,t}),$$
$$\forall g\in\mathcal{G},\forall t\in\mathcal{T}\tag{9e}$$

$$\Big(\sum\nolimits_{d\in\mathcal{D}}p_{d,t,s}-\sum\nolimits_{d\in\mathcal{D}}\Delta p_{d,t,s}\Big)+\sum\nolimits_{i\in\mathcal{I}_b}p_{add,i,t,s}=$$
$$\sum\nolimits_{g\in\mathcal{G}}p_{g,t,s}+\Big(\sum\nolimits_{w\in\mathcal{W}}p_{w,t,s}-\sum\nolimits_{w\in\mathcal{W}}p_{w,t,s}\Big)+$$
$$\Big(\sum\nolimits_{e\in\mathcal{E}}p_{e,t,s}^{dis}-\sum\nolimits_{e\in\mathcal{E}}p_{e,t,s}^{ch}\Big),$$
$$\forall g\in\mathcal{G},\forall w\in\mathcal{W},\forall e\in\mathcal{E},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9f}$$

$$\Big|\sum\nolimits_{b\in\mathcal{B}_l}\iota_{l,b}\Big[\sum\nolimits_{g\in\mathcal{G}_b}p_{g,t,s}-\sum\nolimits_{i\in\mathcal{I}_b}p_{add,i,t,s}+$$
$$\Big(\sum\nolimits_{e\in\mathcal{E}_b}p_{e,t,s}^{dis}-\sum\nolimits_{e\in\mathcal{E}_b}p_{e,t,s}^{ch}\Big)+$$
$$\Big(\sum\nolimits_{w\in\mathcal{W}_b}p_{w,t,s}-\sum\nolimits_{w\in\mathcal{W}_b}\Delta p_{w,t,s}\Big)-$$
$$\Big(\sum\nolimits_{d\in\mathcal{D}_b}\Delta p_{d,t,s}-\sum\nolimits_{d\in\mathcal{D}_b}p_{d,t,s}\Big)\Big]\Big|\le\lceil p_l\rceil,$$
$$\forall l\in\mathcal{L},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9g}$$

$$p_{g,t+1,s}-p_{g,t,s}\le x_{g,t}\gamma_g^{up}+(u_{g,t}-x_{g,t})\,\gamma_g^{su}$$
$$\forall g\in\mathcal{G},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9h}$$

$$p_{g,t,s}-p_{g,t+1,s}\le x_{g,t}\gamma_g^{dw}+(v_{g,t}-x_{g,t})\,\gamma_g^{sd}$$
$$\forall g\in\mathcal{G},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9i}$$

$$\lfloor p_g\rfloor x_{g,t}\le p_{g,t,s}\le\lceil p_g\rceil x_{g,t},\forall g\in\mathcal{G},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9j}$$

$$0\le\Delta p_{w,t,s}\le p_{w,t,s},\quad\forall w\in\mathcal{W},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9k}$$

$$0\le\Delta p_{d,t,s}\le p_{d,t},\quad\forall d\in\mathcal{D},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9l}$$

$$0\le p_{e,t,s}^{ch}\le\lceil p_e^{ch}\rceil\varepsilon_{e,s,t}^{ch},\quad\forall e\in\mathcal{E},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9m}$$

$$0\le p_{e,t,s}^{dis}\le\lceil p_e^{dis}\rceil\varepsilon_{e,s,t}^{dis},\quad\forall e\in\mathcal{E},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9n}$$

$$\varepsilon_{e,s,t}^{ch}+\varepsilon_{e,s,t}^{dis}\le1,\quad\forall e\in\mathcal{E},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9o}$$

$$q_{e,s,t}=q_{e,s,t-1}+\sum\nolimits_{t\in\mathcal{T}}\left[\eta_e^{ch}p_{e,s,t}^{ch}-(1-\eta_{dis})p_{e,s,t}^{dis}\right],$$
$$\forall e\in\mathcal{E},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9p}$$

$$q_{e,s,t}-q_{e,s,t-1}\le\gamma_e^{ch}\varepsilon_{e,s,t}^{ch},\forall e\in\mathcal{E},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9q}$$

$$q_{e,s,t-1}-q_{e,s,t}\le\gamma_e^{dis}\varepsilon_{e,s,t}^{dis},\forall e\in\mathcal{E},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9r}$$

$$cons.(2b)-(2e),cons.(3d)-(3e),cons.(5a)-(5c),$$
$$cons.(6a)-(6h),\quad\forall i\in\mathcal{I},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9s}$$

$$con.(7a),and,con.(8a),\quad\forall i\in\mathcal{I},\forall t\in\mathcal{T},\forall s\in\mathcal{S}\tag{9t}$$

In the above formulations, The objective function (9a) aims to achieve the economic operation cost of the power systems. Constraint (9b) and (9c) enforce the consistency among unit start-up state, shut-down state, and on/off status decisions produced by conventional generators. Constraints (9d) and (9e) impose minimum shut-up/shut-down duration restrictions for each conventional generators, ensuring adequate durations between consecutive start-up and shut-down actions. Constraint (9f) ensures power balance between generation and demand. The demand side includes both conventional electricity consumers and DCCs. Constraint (9g) models the electrical power flow limits on each transmission line. Constraints (9h) enforce the upward and downward ramping limits of individual conventional generators. Constraints (9i) and (9j) specify the upper and lower bounds of power output for each conventional generator. Constraint (9k) restricts the allowable curtailment of wind power, while constraint (9l) accounts for potential forced load curtailment during specific periods. Constraints (9m) and (9n) govern the charging and discharging power of BESS. Constraint (9o) ensures mutual exclusivity between charging and discharging states of BESS, preventing simultaneous charging and discharging. Constraint (9p) tracks the state of charge in BESS, where η_e^{ch} and η_e^{dis} denote the charging and discharging efficiencies, respectively. Constraints (9q) and (9r) define the ramping limits for battery systems during charging and discharging. Constraints (9s) and (9t) represent the physical operating limits of DCCs under DVFS control schemes.

V. Case Studies and Analysis

To validate the effectiveness of the proposed refined UC model, a data center system comprising multiple DCCs is integrated into the IEEE 6-bus test system. The total workload consumption of all DCCs throughout the scheduling horizon is statisticed in Table. 1. Additionally, two wind farms, each

TABLE I
BOUNDARY PARAMETERS OF DCCS

NO.	Bus	$\lfloor p_{dcc,i} \rfloor$	$\lceil p_{dcc,i} \rceil$	$\lambda_{dcc,i}^{ini}$	C_{sv}	$\lambda_{dcc,i}^{dyn}$	$\mu_{dcc,i}$
1	1	0.5	0.10	0.05	0.30	0.50	1.0
2	2	0.8	0.16	0.05	0.30	0.75	1.0
3	6	0.4	0.08	0.05	0.30	0.80	1.0
4	2	0.6	0.12	0.05	0.30	0.30	1.0
5	6	0.5	0.10	0.05	0.30	0.40	1.0
6	2	0.7	0.14	0.05	0.30	0.40	1.0
7	2	0.8	0.16	0.05	0.30	0.60	1.0
8	1	0.5	0.10	0.05	0.30	0.20	1.0

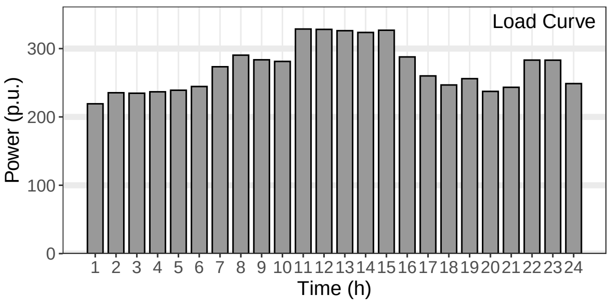

Fig. 2. Chronological load curve of the IEEE 6-bus system with DCCs.

with an installed capacity of 5 MW, are connected at bus 2 in the power system. The sequential load curves and stochastic wind farm outputs are illustrated in Figure []. All experiments are conducted on a personal Lenovo desktop equipped with a Intel(R) Core i7-13700F CPU @3.46GHz and RAM 32GB. Both the proposed model and the benchmark model are implemented in the Julia/JuMP environment. Data visualization is performed using the ggplot2 package in the R programming language.

In this section, the parameters of DCCs referring to [] are set as follows: The idle workload consumption generated by each DCC is set to 10KW, and 8 DCC devices are assumed to be equipped to construct the all workload requirements. Other boundary conditions such as DCC peak power, DCC peak power, workload ratio, initial frequency and supply voltage, DVFS regulation regions, etc., are set as shown in Table. 2.

VI. APPENDIX A: REFORMULATION OF CONSTRAINTS (3C) THROUGH MCCORMICK ENVELOPE METHOD

Let $\gamma_{dcc,i}$ denotes the square value of $v_{dcc,i}^2$, *i.e.*, $\gamma_{dcc,i} = v_{dcc,i}^2$, the lower/upper bound of $\gamma_{dcc,i}$ can be obtained through

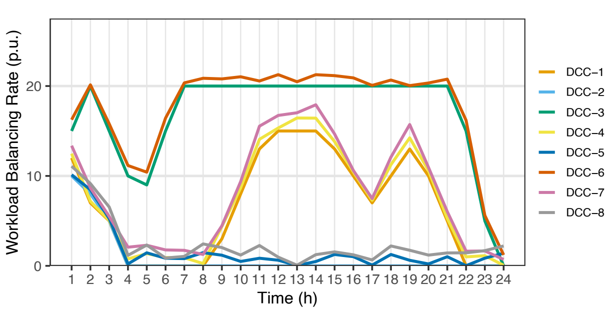

Fig. 3. Given workload distribution of the IEEE 6-bus system with DCCs.

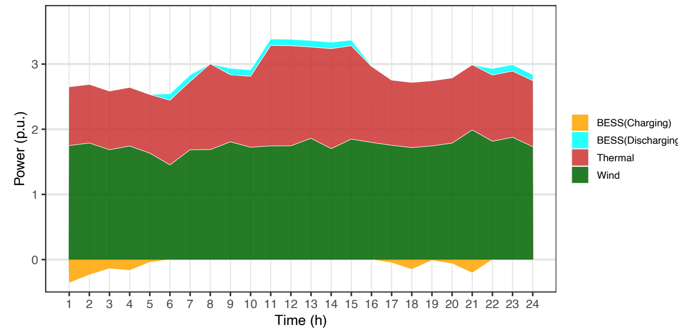

Fig. 4. Power balance process of the IEEE 6-bus system with DCCs.

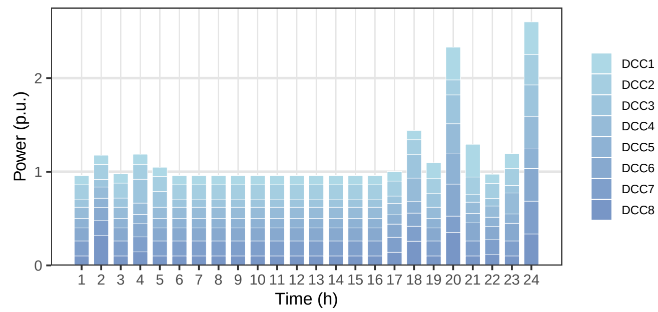

Fig. 5. sequential workload consumption generated by DCCs.

estimating bindings parameters $(\lfloor v_{dcc,i} \rfloor)^2$ and $(\lceil v_{dcc,i} \rceil)^2$, respectively. This process holds because of the monotonic increasing property of the quadratic function. Accordingly, the constraints representing the lower/upper bounds of $\gamma_{dcc,i}$ can be derived as below,

$$\lfloor v_{dcc,i} \rfloor \leq v_{dcc,i} \leq \lceil v_{dcc,i} \rceil, \qquad \forall i \in \mathcal{I} \tag{10a}$$

$$\Rightarrow \quad (\lfloor v_{dcc,i} \rfloor)^2 \leq \gamma_{dcc,i} \leq (\lceil v_{dcc,i} \rceil)^2, \qquad \forall i \in \mathcal{I} \tag{10b}$$

Next, $\phi_{dcc,i}^{dvfs}$ can be equivalently represented as the product of two continuous variables $\gamma_{dcc,i}$ and $f_{dcc,i}$. By utilizing the McCormick Envelope method, a reformulated, linearized approximation of this product, and thus of $\phi_{dcc,i}^{dvfs}$ can be constructed as,

$$\kappa_{dcc,i}^{up} = 1/2(\gamma_{dcc,i} + f_{dcc,i}), \qquad \forall i \in \mathcal{I} \tag{11a}$$

$$\kappa_{dcc,i}^{dw} = 1/2(\gamma_{dcc,i} - f_{dcc,i}), \qquad \forall i \in \mathcal{I} \tag{11b}$$

$$\phi_{dcc,i}^{dvfs} \geq \sum\nolimits_{z \in \mathcal{Z}} \Im(\kappa_{dcc,i,z}^{up}) \vartheta_{dcc,i,z}^{up} - \tag{11c}$$

$$\sum\nolimits_{z \in \mathcal{Z}} \Im(\kappa_{dcc,i,z}^{dw}) \vartheta_{dcc,i,z}^{dw}, \qquad \forall i \in \mathcal{I} \tag{11d}$$

$$\kappa_{dcc,i}^{up} = \sum\nolimits_{z \in \mathcal{Z}} \kappa_{dcc,i,z}^{up} \vartheta_{dcc,i,z}^{up}, \qquad \forall i \in \mathcal{I} \tag{11e}$$

$$\kappa_{dcc,i}^{dw} = \sum\nolimits_{z \in \mathcal{Z}} \kappa_{dcc,i,z}^{dw} \vartheta_{dcc,i,z}^{dw}, \qquad \forall i \in \mathcal{I} \tag{11f}$$

$$\sum\nolimits_{z \in \mathcal{Z}} \vartheta_{dcc,i,z}^{up} = 1, \qquad \forall i \in \mathcal{I} \tag{11g}$$

$$\sum\nolimits_{z \in \mathcal{Z}} \vartheta_{dcc,i,z}^{dw} = 1, \qquad \forall i \in \mathcal{I} \tag{11h}$$

$$\Im(\kappa_{dcc,i,z}^{up}) = (\kappa_{dcc,i,z}^{up})^2, \forall z \in \mathcal{Z}, \qquad \forall i \in \mathcal{I} \tag{11i}$$

$$\Im(\kappa_{dcc,i,z}^{dw}) = (\kappa_{dcc,i,z}^{dw})^2, \forall z \in \mathcal{Z}, \qquad \forall i \in \mathcal{I} \tag{11j}$$

$$\kappa_{dcc,i,z}^{up} \geq 0, \kappa_{dcc,i,z}^{dw} \geq 0, \forall z \in \mathcal{Z}, \qquad \forall i \in \mathcal{I} \tag{11k}$$

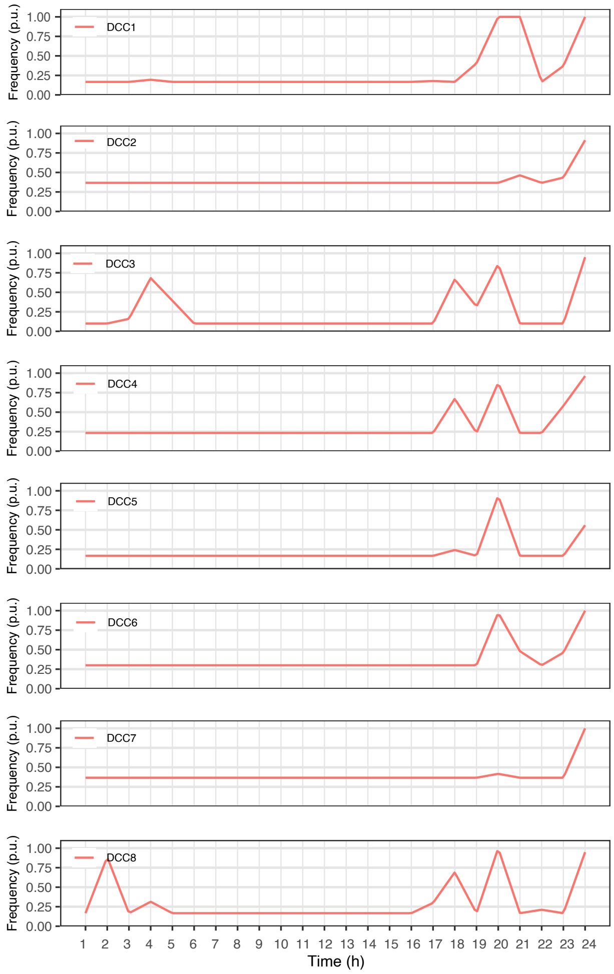

Fig. 6. sequential frequency regulations through DVFS in DCCs.

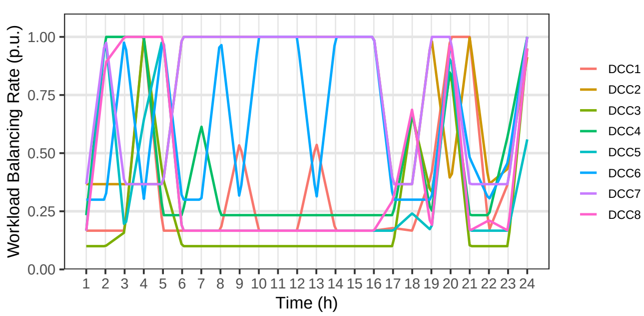

Fig. 7. sequential workload balancing through DVFS in DCCs.

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